Explainable Questionnaire-Based Screening of Anemia and Chronic Energy Deficiency in Pregnant Women: A Case Study in North Lombok, Indonesia

Final Project Proposal

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Skrining Anemia dan Kurang Energi Kronik pada Ibu Hamil Berbasis Kuesioner yang Dapat Dijelaskan: Studi Kasus di Kabupaten Lombok Utara, Indonesia

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Abstract

Anemia affects 27.7 percent of pregnant women in Indonesia according to the 2023 Indonesia Health Survey. Field data from 420 pregnant women in North Lombok Regency, collected for this study, show 21.5 percent below the WHO hemoglobin threshold among the 396 women with a valid hemoglobin value. They also show 27.07 percent at risk of chronic energy deficiency (CED) among the 410 women with a valid arm circumference. Both hemoglobin testing and arm circumference measurement depend on access to a health facility, and many women in the region are not reached.

This study builds an XGBoost regressor that estimates hemoglobin from questionnaire data and an XGBoost classifier that estimates CED risk. Hemoglobin and mid-upper arm circumference (MUAC) are excluded from their own feature sets to prevent target leakage. Feature selection, imputation, and tuning run inside the cross-validation folds. Both models are compared with simple baselines under repeated stratified cross-validation, using feasibility criteria fixed in advance. SHAP TreeExplainer explains individual predictions, and a prototype Streamlit dashboard with SQLite session storage presents the results as a screening support tool. The models are validated internally on one regional dataset, and the dashboard has not been evaluated with community cadres.

Keywords: anemia, chronic energy deficiency, pregnant women, XGBoost, SHAP, screening dashboard, North Lombok.

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Chapter I

Introduction

1.1 Background

Maternal anemia, defined as hemoglobin below 11.0 g/dL, contributes to maternal and infant morbidity and mortality [1]. In Indonesia, prevalence among pregnant women rose from 37.1 percent in 2013 to 48.9 percent in 2018 according to Riskesdas [2]. The 2023 Indonesia Health Survey then reported a fall to 27.7 percent [3]. A national average can hide local differences, which is why this study uses field data from a single regency, North Lombok.

The data were collected from 420 pregnant women in five subdistricts of North Lombok Regency. After cleaning, 396 women have a valid hemoglobin value, and 85 of them (21.5 percent) are below 11.0 g/dL. Of these, 57 are in the mild range and 28 in the moderate range, and no severe case was recorded. Another 16 women have no usable hemoglobin result, and 15 of those 16 said they had never had a blood test during the pregnancy. Among the 410 women with a valid mid-upper arm circumference (MUAC), 111 (27.07 percent) have a recorded value below the 23.5 cm cutoff and are classified as at risk of Chronic Energy Deficiency (CED).

Anemia and CED are usually screened separately. Both are shaped by food access, diet, and use of antenatal care, so one questionnaire can look for both at the same time. Indonesian studies link anemia to iron tablet adherence, parity, nutritional status, gestational age, and employment [4, 5].

Detecting either condition today requires a physical step. Anemia needs a blood test and CED needs a tape measurement, and both depend on reaching a facility with equipment and trained staff. Digital support exists. Veronica et al. built a rule-based web system for early anemia detection among pregnant mothers [6]. Fixed rules, however, cannot learn the patterns that exist in local data.

Machine learning can learn those patterns. Scoping reviews describe machine learning for anemia in pregnancy as promising [1], and one review reports high accuracy for several algorithms, including XGBoost, CatBoost, and Random Forest [7]. Individual studies disagree on the best algorithm. Kitaw et al. compared six models for anemia severity among Ethiopian pregnant women

and found Random Forest the most promising [8]. XGBoost [9] is a natural candidate for tabular survey data, and a large benchmark found tree-based models still strong on medium-sized tabular data of about 10,000 samples [10]. The dataset here is far smaller, so this study does not assume boosting will win. It compares XGBoost with simple baselines and reports the outcome either way.

A flag is only useful if the cadre can see why it was raised. SHAP assigns each feature a Shapley value for an individual prediction [11, 12], which shows which factors pushed a woman's estimate up or down. These values describe the behavior of the model and not causes.

Leakage is the other concern. When hemoglobin or MUAC is used as an input, the model reads the answer from its inputs and performance is inflated [13]. Kapoor and Narayanan found that leakage affected 294 papers across 17 fields [14]. This study removes hemoglobin from the anemia model and MUAC from the CED model, and it fits every data-dependent step inside the training folds.

To the best of our knowledge, we did not find a previous study that estimates hemoglobin and CED risk in parallel from a single questionnaire in an Indonesian community sample, explains the predictions with SHAP, and delivers them through a working dashboard. This study also tests whether a questionnaire carries enough signal at all, which earlier work often assumes. The results come from one regional dataset and are validated internally, so they do not show that the approach generalizes to other regions.

1.2 Research Questions

This study is guided by three questions.

1. Can an XGBoost model trained on questionnaire features screen for anemia (hemoglobin below 11.0 g/dL) with sensitivity of at least 0.70 and specificity of at least 0.40 under leakage-free validation, and does it improve on its baselines?

2. Can an XGBoost classifier reach an AUC-ROC of at least 0.70 for CED risk and outperform its baselines?

3. Can the model outputs and SHAP explanations be delivered in a functional prototype dashboard that accepts questionnaire-only input?

The data are limited to North Lombok Regency. Anemia is analyzed as Normal, Mild, and Moderate, because no severe case was recorded. Hemoglobin was measured at a different time from the questionnaire, and only women with a valid hemoglobin result enter the anemia model. The dashboard is a prototype and no user study with community cadres is part of this work.

Longitudinal follow-up and integration with hospital records are outside the scope.

1.3 Objectives

This study pursues four objectives:

1. To build an XGBoost regressor for hemoglobin and an XGBoost classifier for CED risk from questionnaire features, with feature selection performed inside the cross-validation folds.

2. To compare both models with baselines that are fixed before training.

3. To explain predictions with SHAP and report how stable the explanations are across folds.

4. To implement a functional prototype dashboard with SQLite session storage and test its technical correctness.

The main contribution is the evaluation of questionnaire-based screening with explainable machine learning on one local dataset. The dashboard implements the method as a prototype and is not claimed as a technological innovation or as a tool shown to be effective in the field.

1.4 Hypotheses

This study tests four hypotheses and treats everything else as exploratory. Each hypothesis receives one of three verdicts: supported, not supported, or inconclusive. A verdict is inconclusive when the point estimate and its confidence interval fall on different sides of the criterion. The criteria were fixed before the final experiments, and any later change is recorded with its date and reason. Table 1.1 lists them.

Table 1.1: Hypotheses and success criteria.

Hypothesis	**Criterion**
Primary 1. Anemia screening from predicted hemoglobin	Sensitivity of at least 0.70 and specificity of at least 0.40 on out-of-fold predictions, with the lower bound of the AUC confidence interval above 0.50
Primary 2. CED classification	AUC of at least 0.70 with a confidence interval lower bound of at least 0.50, and at least 0.03 AUC above Logistic Regression
Secondary 1. Hemoglobin error against a mean-only baseline	MAE lower by at least 0.10 g/dL
Secondary 2. Hemoglobin error against Ridge regression	MAE lower by at least 0.05 g/dL

The sensitivity of 0.70 and the specificity of 0.40 are feasibility criteria for this study. They are not clinical standards and no guideline sets them. Section 3.5 explains what they imply for the number of women flagged and missed. Holm correction is applied across the two primary hypotheses. Resampling, direct classification of anemia, and prediction intervals are exploratory and are reported without a verdict.

The dashboard is accepted if the SHAP additivity check passes, the ranking of top features is stable across folds, and the six dashboard test components in Section 3.8 pass.

1.5 Research Activity Plan

The research runs across five phases over six months counted from the approval of this proposal.

1. **Data Preparation.** Apply the cleaning rules in Section 3.2, build both labels, and set up checks that keep hemoglobin and MUAC out of the feature matrices.

2. **Model Development.** Run feature selection inside each fold, tune the XGBoost models and baselines, and fit the SHAP explainers.

3. **Model Evaluation.** Compute metrics, compare each model with its baselines, run the sensitivity analyses, and give a verdict on each hypothesis.

4. **Dashboard Development and Testing.** Build the Streamlit application with SQLite storage and run the test components.

5. **Writing and Submission.** Write the thesis and a conference paper in parallel. Submission is planned at the end of the sixth month.

Table 1.2: Research activity timeline.

No	**Activity**	**1**	**2**	**3**	**4**	**5**	**6**
1	Literature Review						
2	Data Preparation						
3	Model Development and Training						
4	Model Evaluation						
5	Dashboard Development and Testing						
6	Report Writing						

Chapter II

Literature Review

2.1 Related Studies

The related literature falls into four groups: machine learning models for anemia, broader maternal risk prediction, epidemiological studies of anemia in Indonesia, and digital detection tools. Table 2.1 summarizes the closest studies.

Table 2.1: Summary of previous studies.

Study	**Context and Approach**	**Key Limitation for This Work**
Khotimah et al. (2025) [1]	Scoping review of machine learning tools for anemia in pregnancy	Review only, no new model
Kario and Kurniawan (2024) [7]	Scoping review of machine learning for anemia prediction	Review only, no new model
Kitaw et al. (2024) [8]	Six machine learning models for anemia severity in Ethiopian pregnant women	Different country and data source
Nurrahman et al. (2023) [15]	XGBoost for multiclass anemia classification on Riskesdas data	No explainability or deployment reported
Al Mashraf et al. (2024) [16]	Several machine learning models for maternal risk level	Targets maternal risk, not anemia or nutrition
Veronica et al. (2021) [6]	Rule-based web system for early anemia detection	Cannot learn from data
Arfan et al. (2024) [4] and Wahyuningsih et al. (2023) [5]	Statistical studies of anemia risk factors in Indonesia	No predictive model
This Study	XGBoost regressor and classifier with SHAP and a prototype dashboard, North Lombok	Evaluated against baselines under leakage-free validation, internal validation only

The gap is a combination and not a method. XGBoost and SHAP have each been applied to anemia before. What we did not find is parallel estimation of hemoglobin and CED from one questionnaire in an Indonesian community sample, with explanations and a working prototype. The novelty is therefore in the combination of approach and local context and not in a new algorithm.

Using one questionnaire for two targets has a practical benefit, because the cadre fills one form. This study does not claim that estimating both targets together improves accuracy over two separate models, and the two models are trained separately. The literature also gives no reason to expect XGBoost to win, which is why baselines are part of the design.

2.2 Anemia in Pregnant Women in Indonesia

WHO defines anemia in pregnancy as hemoglobin below 11.0 g/dL, with mild (10.0 to 10.9 g/dL), moderate (7.0 to 9.9 g/dL), and severe (below

7.0 g/dL) grades. The 2024 WHO guideline added trimester-specific cutoffs of below 11.0 g/dL in the first trimester (up to 13 weeks), below 10.5 g/dL in the second (14 to 27 weeks), and below 11.0 g/dL in the third (28 weeks onward) [17]. This study uses 11.0 g/dL as the primary threshold, because gestational age was entered inconsistently in the questionnaire. The trimester-specific cutoffs are used as a sensitivity analysis.

Indonesian prevalence was 37.1 percent in 2013, 48.9 percent in 2018 [2], and 27.7 percent in 2023 [3]. Determinants reported in Indonesian studies include iron tablet adherence, parity, nutritional status, gestational age, and employment [4, 5]. Both of those studies are small association studies and are read as such.

2.3 Chronic Energy Deficiency in Pregnant Women

CED in pregnancy is identified by a MUAC below 23.5 cm under Indonesian Ministry of Health standards [2]. MUAC was recorded in whole centimeters in this study, and it is not known whether values were rounded or truncated. A recorded value of 23 or less is therefore treated as below 23.5, and the resulting label is an operational classification called CED risk. It is not a confirmed clinical diagnosis. Section 3.8 tests the neighboring cutoffs of 22 and 24. MUAC defines the CED risk label and is removed from the CED feature set, following the same logic applied to hemoglobin.

2.4 XGBoost

XGBoost builds an additive ensemble of decision trees, where each new tree corrects the residual errors of the trees before it [9]. The final prediction for sample i is

$$\hat{y}_i = \sum_{k=1}^{K} f_k(x_i), \quad f_k \in \mathcal{F} \tag{2.1}$$

where K is the total number of trees and f_k maps inputs to a leaf value from the function space $\mathcal{F}$. Training minimizes the regularized objective

$$\mathcal{L} = \sum_{i=1}^{n} l(y_i, \hat{y}_i) + \sum_{k=1}^{K} \Omega(f_k), \quad \Omega(f_k) = \gamma T_k + \frac{1}{2}\lambda \sum_{j=1}^{T_k} w_j^2 \tag{2.2}$$

The first term measures prediction error: squared error for regression and log-loss for classification. $\Omega(f_k)$ penalizes model complexity through the number of leaves T_k and the squared leaf weights w_j, for $j = 1, \ldots, T_k$, governed by γ and λ. This penalty is what keeps the ensemble from overfitting a small dataset.

This study uses two variants. XGBRegressor with `objective='reg:squarederror'` estimates continuous hemoglobin values. XGBClassifier with `objective='binary:logistic'`

estimates the probability of CED risk and, as an exploratory comparison, the probability of anemia.

Boosting is a candidate here because tree-based models do well on tabular data of moderate size [10]. The samples in this study are below 420, so the baselines answer two questions. A model that predicts only the mean (hemoglobin) or the class proportion (CED) shows whether XGBoost learns anything at all. A regularized linear model, Ridge for hemoglobin and Logistic Regression for CED, shows whether the extra complexity is justified. Random Forest can be added as a third baseline from the same tree family.

2.5 SHAP

SHAP attributes a model output to its input features by computing each feature's Shapley value from cooperative game theory [11]:

$$\phi_i = \sum_{S \subseteq F \setminus \{i\}} \frac{|S|!\,(|F| - |S| - 1)!}{|F|!} \big[f(S \cup \{i\}) - f(S) \big] \tag{2.3}$$

where F is the full feature set, S is a subset excluding feature i, and $f(S)$ is the model output using only features in S. ϕ_i is the average marginal contribution of feature i across all possible feature orderings.

Three properties make SHAP suitable for clinical use. *Local accuracy* means the contributions of one prediction add up exactly to the difference between that prediction and the baseline value. *Missingness* means an absent feature contributes zero. *Consistency* means that if a model change raises a feature's marginal contribution, its SHAP value does not fall [11]. For tree ensembles, TreeSHAP computes exact values in polynomial time [12].

SHAP describes what the model does and not what causes the outcome. It does not show that a prediction is correct, and a model that learned little gives explanations that carry little meaning. Women already known to be anemic may receive more iron tablets, for example, so tablet count could point in the opposite direction from its true effect. The dashboard narratives are written with this limit in mind.

2.6 Class Imbalance and Resampling

The CED risk label has 111 positives and 299 negatives, a ratio of about 1 to 2.7. Anemia has 85 positives among 396 women, and the moderate grade holds only 28 cases. SMOTE creates synthetic minority cases by interpolating between real ones [18], and its SMOTE-NC variant handles mixed numeric and categorical features. SMOGN extends the idea to continuous targets [19]. Class weighting through `scale_pos_weight` is a third option.

Resampling does not always help. In simulations and a case study using logistic regression, every imbalance correction produced poorly calibrated probabilities and none improved AUC [20]. A follow-up simulation covering

machine learning models found that models without a correction had equal or better calibration in every scenario [21]. Based on this evidence, the main models here use no resampling. SMOTE-NC, class weighting, and SMOGN are tested only as exploratory comparisons with the same tuning budget.

2.7 Target Leakage and Validation

Target leakage is the use of information at training time that would not be available at prediction time [13]. Kapoor and Narayanan found it affecting 294 papers across 17 fields [14]. Two sources matter here. The first is using the source of the label as a feature, which is why hemoglobin and MUAC are removed. The second is fitting imputation, feature selection, scaling, or tuning on data that include the evaluation fold. Chapter 3 places all of these steps inside the folds.

Passing a leakage check does not by itself make a model valid for use. A feature can be free of leakage and still be unavailable or unreliable when the tool is used in the field. Chapter 3 treats these as separate questions.

2.8 Uncertainty and Calibration

A hemoglobin estimate is more useful with a range around it. Conformal prediction wraps a trained model and produces intervals that contain the true value with a chosen probability, such as 90 percent [22]. For CED, predicted probabilities are checked against observed frequencies with a reliability curve and the Brier score, and recalibration is tried when they disagree. Both analyses are exploratory.

2.9 Streamlit and SQLite

Streamlit is an open-source Python framework for building data applications without front-end expertise [23]. SQLite is a serverless relational database that stores everything in one local file [24]. Together they keep the system in one language on one machine, which suits a local prototype for primary care settings with limited infrastructure.

Chapter III

Methodology and System Design

3.1 System Architecture

The system has two parts that run independently. The offline pipeline loads the dataset, preprocesses it, trains both XGBoost models and the baselines, fits the SHAP explainers, and saves the trained artifacts. The online prototype dashboard loads those artifacts when a cadre opens it.

The cadre completes one questionnaire. It contains the union of the final features of both models plus the nine food insecurity items, and it has no field for hemoglobin or MUAC. After submission, one inference step runs both models. It returns a hemoglobin estimate with its WHO class and a CED risk probability. The SHAP module then builds a feature attribution chart for each model, and a recommendation module produces educational advice. The recommendation rules are written and edited by a midwife, so the system combines a data-driven model with expert rules. Every session is stored in SQLite, which lets health workers recover past screenings locally without a separate database server. Figure 3.1 shows the architecture.

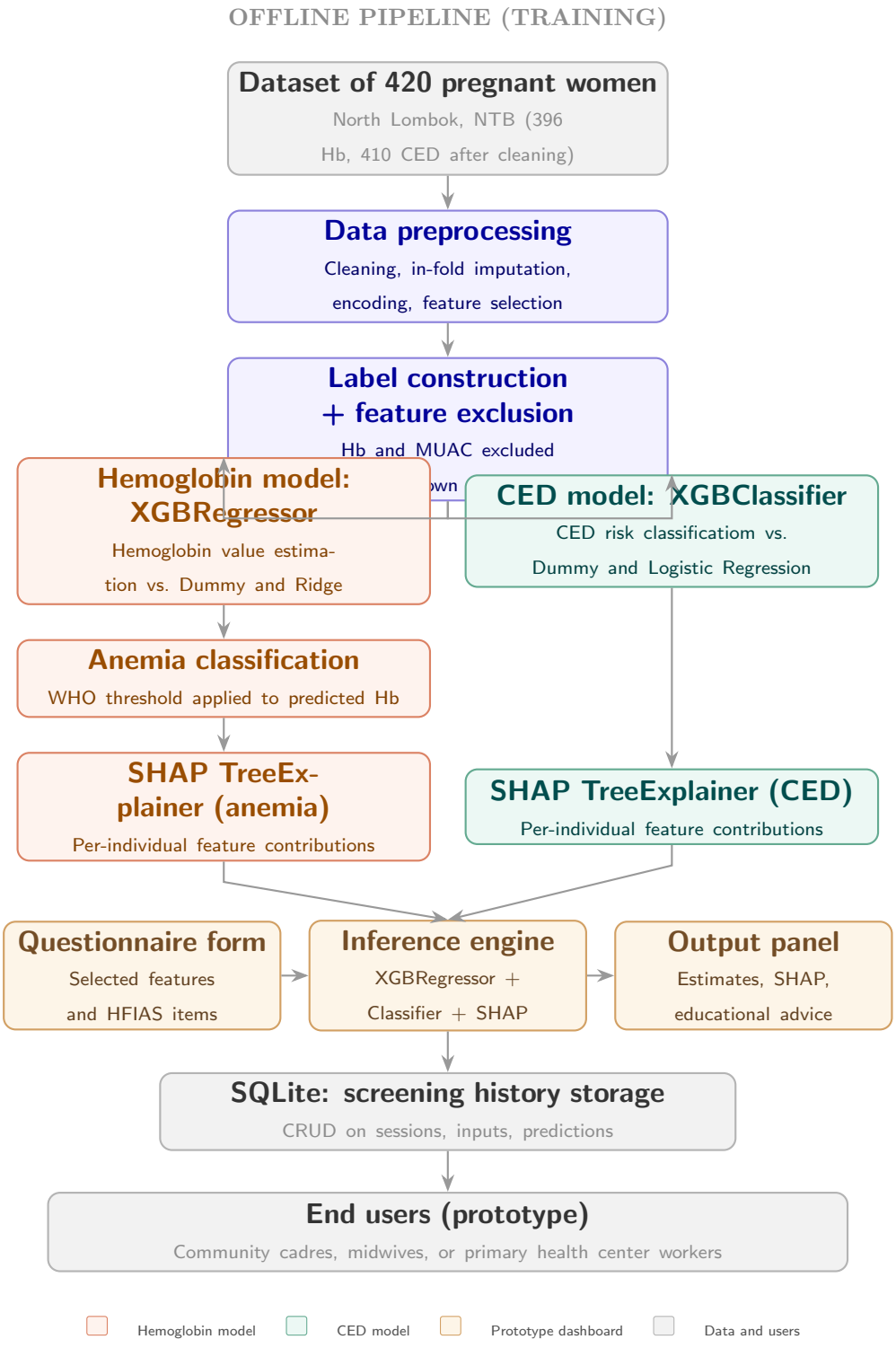


Figure 3.1: System architecture for anemia and CED risk prediction.

3.2 Dataset

The dataset contains 420 records of pregnant women surveyed in five subdistricts of North Lombok Regency, West Nusa Tenggara Province: Gangga, Pemenang, Tanjung, Bayan, and Kayangan. A structured questionnaire covered sociodemographics, obstetric history, anthropometric measurements including hemoglobin and MUAC, diet, iron tablet receipt and adherence, antenatal care, nine food insecurity items, and maternal knowledge. The dataset was provided by the supervising lecturer for use in this thesis, and respondent consent is recorded in the questionnaire. Name, email, phone, and GPS columns are not read into the analysis, and respondents receive anonymous identifiers.

Cleaning follows rules fixed before modeling and is summarized in Table 3.1. One repeated submission from the same respondent is removed, keeping the first. Sixteen records with hemoglobin recorded as 0 are treated as not measured, because 15 of these women said they had never had a blood test during the pregnancy. Seven records with hemoglobin above 20 g/dL (values of 23, 25, 25, 25, 30, 94, and 130) are physiologically implausible and are removed from the anemia model. Some may be decimal point errors, but the original values cannot be verified, so they are not recovered. These 23 records are removed and not imputed, so no label is created by assumption. Women without a hemoglobin value stay in the CED model.

Table 3.1: Cleaning steps and resulting sample sizes.

Step	**Rule**	**Records Remaining**
1	Raw data (Form Responses 1)	420
2	One repeated submission removed	419
3	Hemoglobin equal to 0 treated as not measured (16 records)	403
4	Hemoglobin above 20 g/dL removed from the anemia model (7 records)	**396** (Hb model)
5	MUAC outside 15 to 45 cm removed from the CED model (9 records, applied to the 419 records of step 2)	**410** (CED model)

Plausibility of features. Candidate feature values outside physiologically plausible limits are set to missing and imputed inside the folds. The limits are chosen by the researcher before modeling and are not clinical thresholds: age 15 to 50 years, pre-pregnancy weight 30 to 120 kg, and height 130 to 190 cm. Among the 419 records, one age (3 years), one weight (11 kg), and two heights

(0 and 256 cm) fall outside these ranges. None of the 11 candidate features is otherwise missing, so missingness in the features is not analyzed further.

Missing hemoglobin. The more important gap is missing hemoglobin. The 16 women without a usable value probably have less access to testing, and this is the group a questionnaire-based tool is meant to reach. The anemia model is trained only on women who were tested, so it may not represent women who were not. This is reported as a limitation, and the descriptive profile of these 16 women is reported alongside the results.

With five-fold cross-validation, each training fold holds about 317 records for hemoglobin and 328 for CED. No separate hold-out set is used, because a 20 percent hold-out would contain only about 17 anemia cases, too few for a reliable test.

Anemia labels. Among the 396 records, 311 are Normal (78.54 percent), 57 are Mild (14.39 percent, recorded hemoglobin of 10), and 28 are Moderate (7.07 percent, recorded 8 or 9). No severe case exists. The recorded hemoglobin counts are 8 (2), 9 (26), 10 (57), 11 (106), 12 (138), 13 (55), 14 (10), 15 (1), and 16 (1). The main target is binary, hemoglobin below 11.0 g/dL. A three-class analysis is secondary. Hemoglobin was recorded as a whole number and it is not known whether values were rounded or truncated, so women recorded near the cutoff, especially the 106 recorded at 11, may be misclassified.

CED risk labels. Of the 410 records, 111 (27.07 percent) have a recorded MUAC of 23 cm or less and are labeled at risk. This is an operational label based on recorded MUAC, called CED risk in this thesis, and not a confirmed clinical diagnosis. Of the 111 women at risk, 54 were recorded at exactly 23 cm. If values were truncated, some of them may actually be at or above 23.5 cm. If values were rounded, all of them are below 23.5 cm. The cutoff sensitivity analysis in Section 3.8 addresses both cases.

Prediction target and timing. The hemoglobin target is the value recorded for the woman in the same pregnancy and not her hemoglobin on the interview day. The two were generally measured on different dates. Among the 396 records of the hemoglobin model, 381 have usable dates. For these, the median gap between the hemoglobin date and the interview date is 31 days, 196 records exceed 30 days, and 88 exceed 90 days. Some large gaps come from date entry errors, such as impossible years. The model therefore learns the association between questionnaire answers and a recorded hemoglobin value, and it cannot separate a real change in hemoglobin between the two dates from prediction error. Whether this matters is tested in Section 3.8 with exclusion rules fixed before the final evaluation.

Other fields. The number of children is used as the number of living children and not as parity. The gravida-para-abortus column has inconsistent formats and is used only for validation.

3.3 Feature Selection

Eleven candidate features come from the questionnaire: maternal age, number of children, number of children under five, pre-pregnancy weight, height, ANC visit frequency, food insecurity score, maternal education, household income, egg consumption frequency, and food taboos. Pre-pregnancy weight, ANC frequency, and the food insecurity score are always kept.

The food insecurity score is the sum of the nine HFIAS items [25]. Each item is coded 0 (did not occur), 1 (1 to 2 times), 2 (3 to 10 times), or 3 (more than 10 times), giving a score from 0 to 27. Summing the items avoids feeding nine highly correlated inputs to the model.

Selection runs inside each training fold and not on the full dataset, because selection on the full dataset leaks information into the evaluation. The procedure is fixed as follows. Mutual information (MI) and random forest importance (RF, 300 trees) each rank all 11 candidates from 1 (best) to 11. The two ranks are averaged for each feature. The three features that are always kept are fixed in advance, and the four other features with the lowest average rank complete the set, so each fold uses seven features. The model in each fold is trained on that fold's own selection.

The feature set of the dashboard is defined afterward. A feature enters the form if it was selected in at least half of the folds. The 50 percent threshold is a convention and not a derived value, so the selection frequency of every feature is reported for each model and not only the final list. Two sensitivity setups show whether the selection matters. One uses a fixed seven-feature set from the preliminary analysis, and the other uses all 11 candidate features.

Table 3.2 shows the features selected most consistently in the preliminary analysis on training data. Selection will be rerun on the final 396 and 410 records.

Table 3.2: Features selected in the preliminary analysis.

No	Feature	Hemoglobin Model	CED Model
1	Pre-pregnancy body weight	Yes	Yes
2	ANC visit frequency	Yes	Yes
3	Composite HFIAS food security score	Yes	Yes
4	Mother's age	Yes	Yes
5	Mother's education	Yes	Yes
6	Height	Yes	No
7	Egg consumption frequency	No	Yes
8	Number of children	No	Yes

The dashboard form follows the union of the final features, about eight fields, plus the nine food insecurity items. Pre-pregnancy weight is the strongest predictor of CED. This is not target leakage, because it is a different measurement taken at a different time from MUAC, although the two correlate through nutritional status.

Three questions should be kept apart. The first is technical target leakage, which the exclusion of hemoglobin and MUAC addresses. The second is whether a feature is available when the tool is used. The third is whether the answer is reliable. Pre-pregnancy weight may be recalled and not measured, and ANC frequency depends on records the woman carries. Errors in these answers reduce the performance of a deployed tool even when no leakage exists, and this is reported as a limitation.

3.4 Data Preprocessing

Every step below is fit on the training portion of a fold only and then applied to the held-out portion.

1. **Leakage check**
   A check confirms that hemoglobin and MUAC are absent from the feature matrix.

2. **Implausible values**
   Values outside the limits in Section 3.2 become missing.

3. **Missing value imputation**
   Numeric columns receive median imputation and categorical columns receive mode imputation. The median is preferred because the anthropometric distributions contain outliers.

4. **Feature selection**
   Features are selected as described in Section 3.3.

5. **Feature encoding**
   Education and income carry a natural ordering and use ordinal encoding. Nominal variables use one-hot encoding, so the model does not read false numerical distances between unordered categories.

6. **Feature scaling**
   Scaling is applied only for Ridge and Logistic Regression, because tree models do not depend on feature scale.

7. **Second leakage check**
   The same check runs again before the model is fit.

3.5 XGBoost Modeling

Two XGBoost models are trained independently. Anemia and CED are clinically different conditions despite shared risk factors, and separate training allows separate tuning and cleaner SHAP attributions for each target. The model structure follows Equations 2.1 and 2.2.

- **Hemoglobin prediction model (XGBRegressor)**
  Trained with `objective='reg:squarederror'`. The output is a continuous hemoglobin estimate in g/dL, which is converted to a WHO class by threshold. The estimate also shows how far a woman sits from the cutoff, which a direct classifier cannot do. Baselines are a mean-only model and Ridge regression.

- **CED risk model (XGBClassifier)**
  Trained with `objective='binary:logistic'`. The output is the probability of CED risk. Baselines are a class-proportion model and Logistic Regression. Random Forest is an optional baseline for both models.

- **Validation and tuning**
  Each model is evaluated with repeated stratified 5 by 5 cross-validation, which is the outer loop. Inside every outer training set, an inner three-fold cross-validation is used for two purposes: tuning the hyperparameters of XGBoost with Optuna [26] and producing the predictions used to set the decision threshold. Imputation, feature selection, and scaling are part of the pipeline and are refit within every inner training split. The outer test fold is never used for any of these choices. Every tuned model receives the same tuning budget, and the regularization strengths of Ridge and Logistic Regression are tuned the same way.

- **Stratification rule**
  Folds are stratified on the combination of anemia class and CED status. A group with fewer than five records is merged into the group with the same anemia class and the opposite CED status. The rule is applied before any splitting, so the folds are reproducible.

- **Decision thresholds**
  The cutoff that turns a hemoglobin estimate into an anemia flag is set from the inner-fold predictions of the outer training set, at the value where 70 percent of the true anemia cases in those predictions are flagged. It is then applied unchanged to the outer test fold. Predictions of a model on its own training data are not used, because they are more accurate than predictions for new women. Choosing a cutoff on the same predictions that are later scored would overstate performance. The CED threshold is chosen the same way using Youden's index. Youden's index treats a missed case and a false alarm as equally costly, which may not hold clinically, so CED results at higher-sensitivity thresholds are also reported.

- **Meaning of the screening criteria**
  The sensitivity of 0.70 and specificity of 0.40 are feasibility criteria for this study. They are not clinical standards. At a prevalence of 21.5 percent, a model that exactly meets them flags about 62 of every 100 women, identifies about 15 of the 21.5 anemic women, and misses about 6.5. Its positive predictive value is about 24 percent and its negative predictive value about 83 percent. Confusion matrices, predictive values, and the referral burden are therefore reported alongside every result.

- **Exploratory comparisons**
  For CED, the main model is compared with SMOTE-NC and class weighting. For hemoglobin, it is compared with SMOGN. An XGBClassifier trained directly on the binary anemia label (hemoglobin below 11.0 g/dL) is compared with regression followed by a threshold, using the same features, folds, and tuning budget. Each variant is retuned with the same budget. Split-conformal intervals with 90 percent nominal coverage are computed for hemoglobin, and reliability curves are drawn for CED. None of these comparisons receives a verdict.

3.6 SHAP Integration

A TreeExplainer is fitted for each final model, using the Shapley formulation in Equation 2.3. Two levels of analysis are produced. Global analysis averages absolute SHAP values and compares the resulting ranking with the built-in XGBoost importance using Spearman rank correlation. Local analysis

produces a waterfall plot for each prediction with a short plain-language narrative in Indonesian. An additivity test checks that the SHAP contributions plus the base value reproduce each model output.

Stability of the explanation is measured as the mean pairwise Spearman correlation of mean absolute SHAP values across folds, reported descriptively without a fixed pass mark. The stability of individual explanations is checked by how often the top three features of a woman stay the same when the model is trained on different folds.

SHAP explains how the model reached an output. It does not show that a feature causes anemia or CED. The hemoglobin explanation is shown in the dashboard only if the hemoglobin model improves on the mean-only baseline by the margin in Table 1.1, because explaining a model that learned little would be misleading. Three things are kept separate. Model explanation says which features raised or lowered an estimate. Educational advice is written from guidelines and reviewed by a health worker. Clinical decisions remain with health workers. Only the first is produced by the model.

3.7 Dashboard Design

The dashboard is a functional prototype built on Streamlit [23] with SQLite as the storage layer [24]. The *presentation layer* shows the form, results, SHAP plots, and educational advice. The *logic layer* loads both models and explainers, repeats the training-time preprocessing on new inputs, and maps outputs and dominant SHAP features to advice. The *data layer* holds three tables. `sessions` stores a session identifier, timestamp, and anonymous user identifier. `inputs` stores the submitted answers, with self-reported fields labeled as such. `predictions` stores the hemoglobin estimate, WHO class, CED risk probability, and top SHAP contributors.

The main page has four panels. The *input form* has no field for hemoglobin or MUAC. The *prediction panel* shows the hemoglobin estimate with its range and WHO class, and the CED risk probability. The *explanation panel* shows a SHAP waterfall plot for each model with a plain-language narrative. The *advice panel* gives educational suggestions tied to the dominant factors, such as dietary access when food insecurity drives a prediction or visit scheduling when ANC frequency does. A session history supports create, read, update, and delete operations.

Every output screen states that the result is an estimate for initial screening support and not a diagnosis, and that it does not replace a blood test or a MUAC measurement. How the hemoglobin result is presented depends on the evaluation. If the anemia model does not meet its criterion, the hemoglobin output is labeled as indicative and the CED risk output is presented as the main result.

Evaluation with users. No user study with community cadres is part of

this thesis. Claims about the dashboard are therefore limited to technical correctness and consistency with the offline models. Whether cadres understand the estimates, the SHAP narratives, and the limits of the tool is left for future work.

Data protection. The prototype stores sessions in a local SQLite file on the device where it runs. It stores anonymous identifiers only and never names, phone numbers, or coordinates. Access depends on the operating system account, and the prototype has no user authentication, so it is not to be deployed on shared devices in this form. Test records are deleted when testing ends.

3.8 Evaluation Protocol

1. **Model performance**
   The hemoglobin model is evaluated with MAE, RMSE, and R^2, each with bootstrap confidence intervals. After applying the WHO cutoff, anemia screening is evaluated with sensitivity, specificity, predictive values, AUC-ROC, and PR-AUC on out-of-fold predictions. The CED model is evaluated with the same classification metrics plus the Brier score. Confusion matrices are examined for both models, with attention to the anemia and CED groups because they carry the highest clinical consequence.

   For each repetition, the predictions from the five outer test folds are pooled, so every woman has one out-of-fold prediction. Metrics are computed on the pooled predictions. The 95 percent confidence intervals are percentile bootstrap intervals over women with 2,000 resamples, and reported values are averages over the five repetitions. A comparison between two models uses the same resampled women for both models, which gives a paired interval.

   Sensitivity is estimated from only 85 anemia cases, so a point estimate of 0.70 has a 95 percent confidence interval of roughly 0.60 to 0.80. This width is the reason a verdict can be inconclusive.

2. **Baseline comparison**
   Each XGBoost model is compared with its baselines under identical preprocessing and identical folds. Repeated cross-validation reuses the same records, so the corrected resampled t-test [27] is used only as an approximate adjustment. Each comparison is also reported as a paired difference with a bootstrap confidence interval, and conclusions rest on the interval. Holm adjustment [28] is applied across the two primary hypotheses. The comparison also examines which features each algorithm prioritizes and whether those rankings match the epidemiological literature on anemia in Indonesia [4, 5].

3. **Sensitivity analyses**
   Six checks test whether conclusions depend on analysis choices:

   - CED cutoffs of 22 cm and 24 cm in addition to 23 cm.
   - The trimester-specific hemoglobin cutoffs from Section 2.2.
   - Exclusion of hemoglobin records whose date differs from the interview date by more than 90 days, and separately by more than 30 days. Records without a usable date are excluded in these two analyses, and sample sizes and performance after exclusion are reported.
   - Cross-validation grouped by subdistrict, to check that results hold across areas.
   - A model using all 11 candidate features, and a model using the fixed seven-feature set.
   - A three-class anemia analysis.

4. **Interpretation limits**
   All results are internal validation on one regional dataset of fewer than 420 women. They show whether the approach is feasible in this dataset and do not show that it generalizes to other regions or that it is clinically useful. A result that does not meet a criterion is reported as such.

5. **Dashboard testing**
   Six components are tested: input validation against out-of-range values, backend output against offline predictions for identical inputs, SHAP plots against direct TreeExplainer output, educational advice against manually built risk profiles, database integrity across multiple sessions, and layout on desktop and mobile screens. Automated checks also confirm that no hemoglobin or MUAC column reaches the feature matrix and that no preprocessing step is fit on evaluation data.

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Appendix